Statistical APT: Improvements and Pitfalls

Sign Indeterminacy, Volatility Heterogeneity and Universe Changes

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 8

A statistical APT model estimates the loadings and factors from the $n \times T$ matrix of returns $\mathbf{R}$ by PCA, re-estimated as the window rolls forward. Three practical problems arise: the signs of the factors are arbitrary, high-volatility stocks dominate the factors, and the universe of stocks changes.

1 Sign Indeterminacy

Problem. If $\mathbf{u}$ is a singular vector of $\mathbf{R}$, so is $-\mathbf{u}$. When the model is re-estimated each day, the columns of the loading matrix $\mathbf{B}_t$ can flip sign (or, when singular values are close, rotate) from one day to the next, so the factor return series are discontinuous.

Procrustes rotation. Align today's loadings with yesterday's, computed on the same stocks, by the orthogonal $k \times k$ matrix

$$\mathbf{Q}_t = \arg \min_{\mathbf{Q}^T\mathbf{Q}=\mathbf{I}_k} \|\mathbf{B}_t\mathbf{Q} - \mathbf{B}_{t-1}\|_F^2$$

If $\mathbf{B}_t^T\mathbf{B}_{t-1} = \mathbf{P}\mathbf{\Lambda}\mathbf{W}^T$ is an SVD, then $\mathbf{Q}_t = \mathbf{P}\mathbf{W}^T$. The aligned loadings and factors are

$$\tilde{\mathbf{B}}_t = \mathbf{B}_t\mathbf{Q}_t$$

$$\tilde{\mathbf{f}}_t = \mathbf{Q}_t^T\mathbf{f}_t$$

and $\tilde{\mathbf{B}}_t\tilde{\mathbf{f}}_t = \mathbf{B}_t\mathbf{f}_t$, so the fitted returns and the covariance estimate are unchanged.

2 Volatility Heterogeneity and Two-Stage PCA

Problem. PCA maximizes explained variance, so stocks with large idiosyncratic variance pull the leading singular vectors toward themselves, and the estimated factors describe a few volatile stocks rather than common variation.

Stage 1. Compute the SVD $\mathbf{R} = \mathbf{U}\mathbf{S}\mathbf{V}^T$, the residuals of the first k factors

$$\mathbf{E} = \mathbf{R} - \mathbf{U}_k\mathbf{S}_k\mathbf{V}_k^T$$

and the idiosyncratic variances

$$\hat{\sigma}_i^2 = \frac{1}{T}\sum_{t=1}^{T} E_{it}^2$$

Stage 2. Scale each stock by its idiosyncratic volatility, $\mathbf{W} = \mathrm{diag}(1/\hat{\sigma}_1, \ldots, 1/\hat{\sigma}_n)$, and compute the SVD

$$\mathbf{W}\mathbf{R} = \tilde{\mathbf{U}}\tilde{\mathbf{S}}\tilde{\mathbf{V}}^T$$

The loadings in the original units and the factor realizations are

$$\mathbf{B} = \mathbf{W}^{-1}\tilde{\mathbf{U}}_k$$

$$\mathbf{F} = \tilde{\mathbf{S}}_k\tilde{\mathbf{V}}_k^T$$

so that $\mathbf{R} \approx \mathbf{B}\mathbf{F}$. Each stock now has roughly equal influence on the factors, as in generalized least squares. In practice, floor $\hat{\sigma}_i$ to avoid dividing by very small values. PCA on the correlation matrix is a simpler alternative, but it scales by total rather than idiosyncratic volatility.

3 Universe Changes

Problem. Stocks enter and leave the universe through listings, delistings, index changes and liquidity filters. The model must give loadings for new stocks and keep factor definitions comparable over time.

3.1 Statistical Models

Both the loadings and the factors are estimated from the returns of the current universe. A new stock has no loading until it has a return history, and a large change in composition can change what the leading factors represent. This is the hardest case. Remedies:

- Re-estimate each period on the stocks with enough history, and align with the previous period by Procrustes rotation on the stocks common to both.
- For a new stock i with a short history, estimate its loadings by regressing its returns on the factor realizations, $\mathbf{r}_i = \mathbf{F}^T\mathbf{b}_i + \mathbf{e}_i$, over the dates available.
- Alternatively, start new stocks with exposures from characteristics and switch to statistical loadings as history accumulates.

3.2 Economic Models

The factors are observed series (market return, interest rate changes, oil prices), so they do not depend on the universe and keep their meaning over time. The loadings of each stock are estimated by a time-series regression of its returns on the factors. A new stock needs enough return history for this regression; a deleted stock is simply dropped.

3.3 Fundamental Models

The exposures are observed characteristics (size, book-to-market, and so on), and the factor returns are estimated by a cross-sectional regression of returns on the exposures each period. A new stock has exposures as soon as its characteristics are known, a deleted stock drops out of the next regression, and the factors keep their meaning. The cost is that characteristics must be available for every stock.

	Fundamental	**Economic**	**Statistical**
Factor continuity	High	Very high	Low
New stocks	Easy	Medium	Hard
Deletions	Easy	Easy	Medium
External data needed	Yes	Yes	No
Interpretability	High	High	Low

Table 1: APT model types and changes in the universe.